

Technology Stack

Date	7 Jun 2026
Team ID	xxxxxx
Project Name	Smart Crop
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	React.js, Tailwind CSS	Enables a highly responsive, component-driven user interface. This ensures that farmers entering soil details on mobile devices experience fast, intuitive interfaces, while researchers get dynamic dashboards for macro analysis graphs.
2	Backend / Server-Side	Python / FastAPI	Chosen for high performance, rapid processing of multi-variable calculations, and seamless native execution of analytical script models. Essential for processing N-P-K, pH, and climate variables instantly.
3	Database / Data Storage	PostgreSQL	Provides robust relational mapping for complex crop threshold tables, soil metric bounds, and structured historic trend data. Offers advanced data integrity and geospatial extensions helpful for agricultural mapping.

4	Cloud / Hosting / Deployment	AWS (EC2, RDS) & Docker	<i>Containerization via Docker ensures environments remain perfectly consistent across development and production. AWS provides reliable, cost-effective infrastructure and easy horizontal scaling during heavy seasonal planting workloads.</i>
5	Version Control & CI/CD	GitHub & GitHub Actions	<i>Facilitates smooth team collaboration, systematic codebase tracking, and automated unit testing workflows to verify engine mathematics before production deployment.</i>
6	Third-Party APIs / Other Tools	OpenWeatherMap API, Google Maps Platform API	The weather API delivers real-time historical and live localized precipitation, humidity, and temperature data. The maps layer allows regional policymakers to visualize agricultural production trends geographically.